

MITHIL ASTIK

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Education

BITS Pilani, Hyderabad Campus

MSc. Physics and B.E. Electrical and Electronics; CGPA: 6.98

Hyderabad, TS

Aug. 2024 – Present

Gujarat Public School, Atladra

Higher-Secondary Education; Grade: 86%

Vadodara, GJ

Jun. 2022 – Apr. 2024

Experience

Freelance Full-Stack Developer & Designer

Feb. 2026 – Present

Multiple Clients

Remote

- Built a full-stack inventory management web app to track phone models and streamline daily HQ-to-branch ordering for a retail client, using Next.js, PostgreSQL, and TypeScript.
- Designed a portfolio website for a 3D designer client, and produced 30+ event posters in two nights for the BPHC cultural fest using Photoshop and Illustrator.

Summer Intern

May 2026 – July 2026

Beijan

BITS Pilani, Hyderabad, TS

- Developed a custom React-based analytics tool to parse, process, and visualize hardware telemetry data, and configured edge-computing interfaces for GPS-denied navigation.
- Spearheaded a digital presence overhaul: designed a mobile-first website and architected an automated multi-platform outreach pipeline.

Quantum Computing Team Lead (prev. Team Member)

Feb. 2025 – Present

IEEE Student Branch, BITS Pilani

Hyderabad, TS

- Developed applied quantum algorithms (e.g., "Quantum Tic-Tac-Toe") using Python and Qiskit; completed the Qiskit Global Summer School and IBM Quantum Learning Modules.
- Secured an IBM Quantum-sponsored partnership and directed the full life-cycle of Quantumverse Vol. 3 (Qiskit Fall Fest 2025), integrating the workshop into ATMoS '25.

Projects

ML Attack Detection for TF-QKD | Python, XGBoost, PyTorch, Qiskit, Docker, MLflow

2026

- Built an ML anomaly-detection framework for continuous-wave side-channel attacks (FIM, TWIRL) on Optical Injection Locking-based Twin-Field QKD, simulating Lang-Kobayashi rate equations to generate training data.
- Achieved 87.53% accuracy with a class-weighted XGBoost classifier over SVM/FedAvg baselines; used SHAP for tiered, explainable alerts.

FraudGuard | Python, FastAPI, XGBoost, LightGBM, SHAP, SQLite, MLflow, Docker

June 2026

- Built an end-to-end ML system scoring financial transactions for fraud risk in real time via FastAPI and a live React dashboard.
- Selected a class-weighted XGBoost model (PR-AUC ~0.96); added cost-aware decisioning and SHAP explainability.

Agentic AI Personal Assistant | Python, Node.js, Electron, SQLite

June 2026 - Present

- Architected a Cloud Brain / Local Hands agentic AI assistant with an always-on-top Electron HUD and Gemini Multimodal Live API integration.
- Built a Python automation daemon for mouse/keyboard control and UI inspection, plus vector memory and a 15+ tool function-calling router.

Webcam Gesture Music Instrument | JavaScript, MediaPipe, Tone.js, WebAudio

2026

- Built a browser-based "air instrument" using real-time MediaPipe hand tracking mapped to Tone.js synthesis, with one-euro-filtered landmarks and scale-quantized gesture-to-note mapping for sub-80ms latency.

P2P Offline Chat App | Flutter, Rust, MongoDB

June 2026 - Present

- Built an E2E-encrypted, peer-to-peer messaging app (WhatsApp-style) that communicates over local networks without requiring internet access.

Beijan Website Redesign | Next.js, Tailwind CSS, Three.js

June - July 2026

- Redesigned and rebuilt beijan.com end-to-end using Next.js and Tailwind CSS with interactive Three.js 3D elements and optimized SEO, improving brand presentation and search visibility.

Technical Skills

Languages: Python, C/C++, Rust, TypeScript, SQL (Postgres, SQLite), JavaScript, HTML/CSS

Frameworks: React, Node.js, Next.js, Express, FastAPI, Flutter, Tailwind CSS, WordPress

Developer Tools & MLOps: Git, Docker, MLflow, Anaconda, GCP, MongoDB, Supabase, Firebase, VS Code, Visual Studio, PyCharm

Libraries: Pandas, NumPy, Matplotlib, Scikit-learn, Seaborn, TensorFlow, PyTorch, XGBoost, LightGBM, SHAP, Qiskit, PennyLane

Softwares: Adobe Photoshop, Adobe Illustrator, Affinity Designer, Figma, Canva, ANSYS Lumerical, Microsoft Office